Practical 19: Fetch and Parse JSON from Web

Source: MODULE 2.pdf | Source pages: 72–79

Aim

To develop and test an Android application demonstrating fetch and parse json from web.

Steps to Perform the Practical

1. Create a project for the JSON networking practical.

2. Review the supplied URL and JSON response structure in the source code.

3. Add the INTERNET permission to AndroidManifest.xml if required.

4. Use the supplied networking code to fetch the JSON data away from the main UI thread.

5. Parse the response and display the required fields in the app.

6. Run the app with internet access and verify that the fetched data appears. Check Logcat if a request fails.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

12. Develop an Android application to fetch and parse JSON data from a web

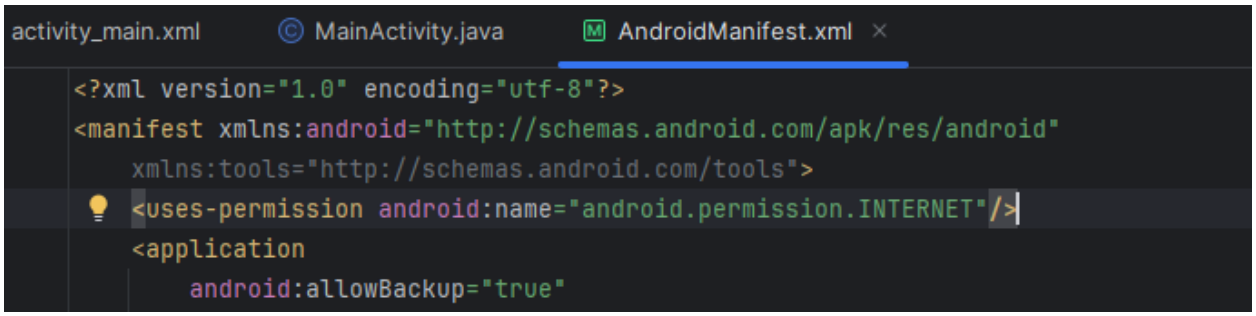
PROJECTNAME: JSONDEMO

Add Internet Permission.

AndroidManifest.xml

Add following above: <application> tag

<uses-permission android:name="android.permission.INTERNET"/>



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:padding="20dp">



   <Button

       android:id="@+id/btnFetch"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"
```

```
        android:text="Fetch JSON Data" />


    <TextView
        android:id="@+id/txtResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:textSize="18sp"
        android:textColor="#000000" />


</LinearLayout>
```

MainActivity.java

package com.example.jsondemo;

```
package com.example.jsondemo;


import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.net.HttpURLConnection;
import java.net.URL;
import org.json.JSONObject;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {



    Button btnFetch;
    TextView txtResult;



    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);



        setContentView(R.layout.activity_main);



        btnFetch = findViewById(R.id.btnFetch);
        txtResult = findViewById(R.id.txtResult);



        btnFetch.setOnClickListener(v -> {
            fetchData();
        });
    }
```

```
   private void fetchData() {



       new Thread(() -> {



           try {



               // Web Service URL

               URL url = new
URL("https://jsonplaceholder.typicode.com/posts/1");



               HttpURLConnection connection =

                       (HttpURLConnection) url.openConnection();


               connection.setRequestMethod("GET");



               connection.setRequestMethod("GET");



               BufferedReader reader = new BufferedReader(new
InputStreamReader(

                               connection.getInputStream()
```

```
        )
);



StringBuilder result = new StringBuilder();



String line;



while ((line = reader.readLine()) != null) {
    result.append(line);
}



reader.close();



// JSON Parsing
JSONObject jsonObject =
        new JSONObject(result.toString());



String title =
        jsonObject.getString("title");
```

```
            String body =
                    jsonObject.getString("body");



            // Display Data
            runOnUiThread(() -> {



                txtResult.setText(
                        "Title: " + title +
                                "\n\nBody: " + body
                );
            });


            connection.disconnect();


        } catch (Exception e) {
            e.printStackTrace();
            runOnUiThread(() -> {
                txtResult.setText(
                        "Error: " + e.getMessage()
                );
            });
        }
```

```
        }).start();

    }

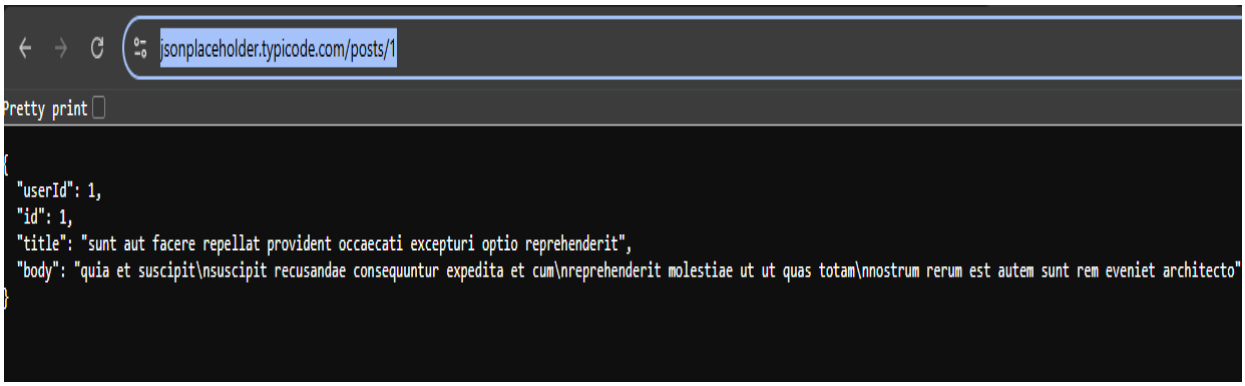
}
```

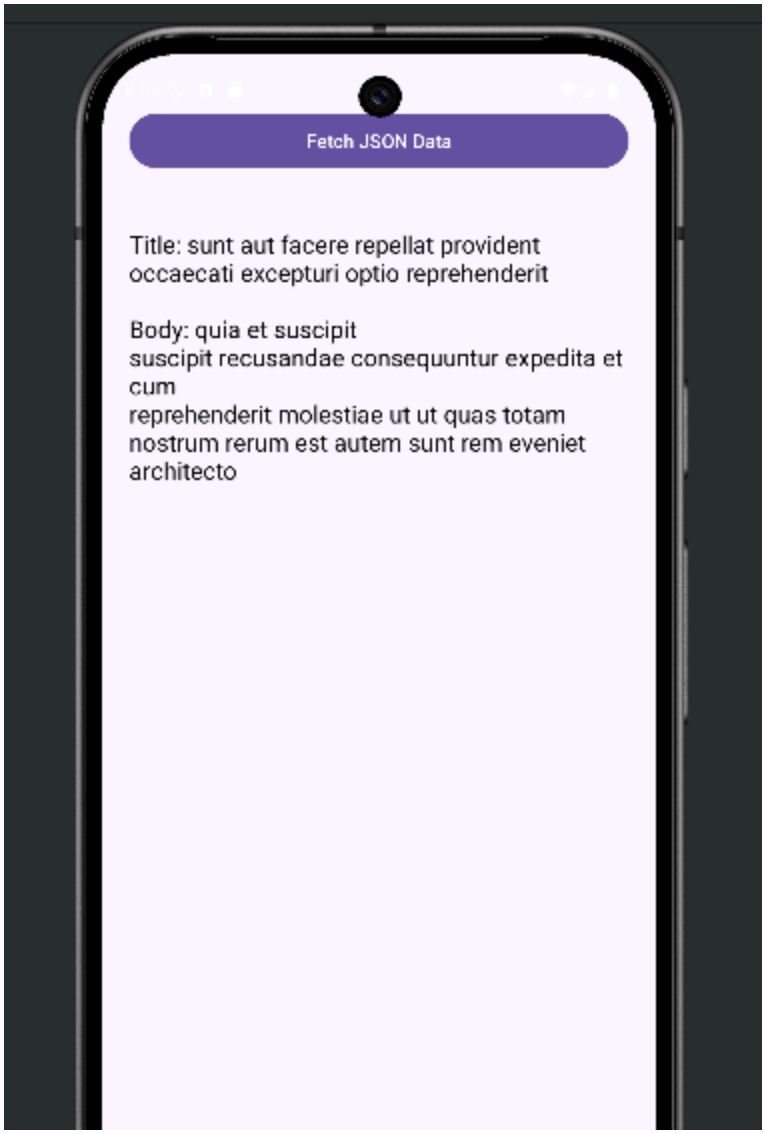
Step 6 : Web Service Used.

API URL : ***https://jsonplaceholder.typicode.com/posts/1***

This is a free test API for students.

https://jsonplaceholder.typicode.com/posts/1




Fetch JSON Data
Title: sunt aut facere repellat provident occaecati excepturi optio reprehenderit
Body: quia et suscipit
suscipit recusandae consequuntur expedita et cum
reprehenderit molestiae ut ut quas totam
nostrum rerum est autem sunt rem eveniet architecto